FIRST PRE-DEFENSE / 24 OCT 2026

Business Management & CRM Platform

with an Embedded AI Assistant

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PROBLEM DEFINITION

Why this project?

An engineering hypothesis that will be checked with real service-business users

Fragmented client data

Customer profiles and inquiries may be stored in disconnected spreadsheets or tools.

Unlinked follow-ups

Deals, task assignments and service appointments can lose shared context.

Booking conflicts

Parallel scheduling needs a consistent server-side rule to prevent overlaps.

Uncontrolled AI actions

Language-generated suggestions need permission checks and explicit confirmation.

RESEARCH STATUS These are hypotheses; authentic interviews/client evidence have not yet been supplied.

One coherent service-business workflow

Trace a client request through the organization, from intake to reporting

01 **Inquiry** › 02 **Client** › 03 **Deal** › 04 **Task** › 05 **Booking** › 06 **Report**

AI assistant: draft → show exact changes → confirm → enforce server rules

First design assumption: one service business, staff-facing web app; no customer mobile app in the MVP.

Planned MVP: six connected capabilities

01

Clients & notes

Create, search and update customer records.

02

Deal pipeline

Track opportunities, stages and owners.

03

Tasks

Assign follow-up work and due dates.

04

Service bookings

Select staff and reject conflicting time slots.

05

Reports

Summarize counts and process outcomes.

06

AI assistant

Prepare typed drafts; confirm safe actions.

A strict MVP keeps the project achievable

IN SCOPE

- ✓ Role-based staff access
- ✓ Clients, deals, tasks and appointments
- ✓ Basic reports and activity history
- ✓ AI action preview and confirmation
- ✓ Transactional booking constraints

OUT OF SCOPE

- – Payroll, accounting, payment processing
- – Full multi-branch ERP and mobile apps
- – Autonomous customer messaging
- – Unreviewed AI database writes
- – Enterprise-scale rollout commitments

Validation: document real user needs

University requirement: 3–5 discussions or one substantive real-client agreement

01

Recruit

Owners, managers, scheduling staff

02

Discuss

Current tools, delays, conflicts and exceptions

03

Record

Consent, dated notes, anonymized quotations

04

Validate

Distinguish observed needs from our hypotheses

NOT YET VERIFIED Templates prepared; real interviews or client confirmation must be collected by the team.

13 scholarly publications inform the design

12 publications (2021–24) plus one foundational reference (2020)

Customer workflows

AI in marketing and service

Huang & Rust · Ferraro et al.

Tool-using models

ReAct, Toolformer, ToolLLM, API-Bank

Yao · Schick · Qin · Li

Failures & security

Prompt injection, lost context, evaluation

Greshake · Liu · AgentBench

Controlled implementation

Augmented models, feedback, retrieval

Mialon · Shinn · Lewis · Wei

Existing systems: compare, do not dismiss

Four vendors assessed against nine documented criteria

HubSpot	Broad CRM, sales, meeting and AI functionality
Zoho CRM	CRM and Zia intelligent-assistance features
Pipedrive	Pipeline and activity scheduling, with conflict avoidance
Bitrix24	CRM + tasks, calendars, analytics and CoPilot

Our engineering focus

01 **End-to-end traceability**
A request, task and booking are linked.

02 **Testable constraints**
No accepted overlapping confirmed bookings.

03 **Auditable AI actions**
Draft, approval and server-side checks.

Technology decisions: compare the alternatives

Selections are justified design proposals, not benchmark results

Backend	**Go**	NestJS · Spring Boot
Storage	**PostgreSQL**	MySQL · MongoDB
API	**REST + OpenAPI**	GraphQL · gRPC
Frontend	**React + TypeScript**	Vue · Angular
AI operations	**Typed proposal + review**	Rule-based · direct LLM execution[1]

[1] Direct model-to-database execution considered and rejected for security; not a proposed implementation.

SPECIFICATION

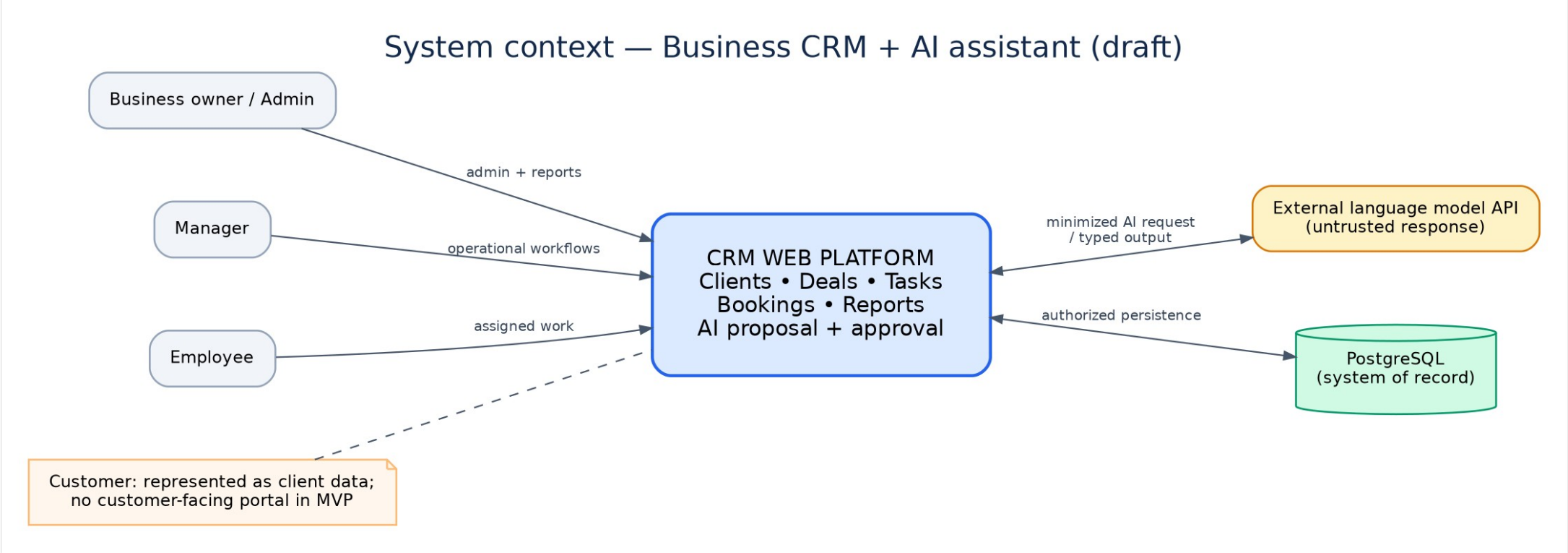
Requirements are measurable

22 functional requirements + 12 non-functional requirements with priorities and acceptance criteria

FR-07	**Book service**	A valid free slot persists with staff, service and client links.
FR-08	**Prevent collisions**	Concurrent overlapping confirmations never create two accepted bookings.
FR-12	**AI proposal**	Natural-language request produces a preview; database is unchanged.
FR-14	**Confirm AI action**	Revalidate role and business rules before any mutation.
NFR-01	**Authorization**	All protected operations deny unauthorized changes.

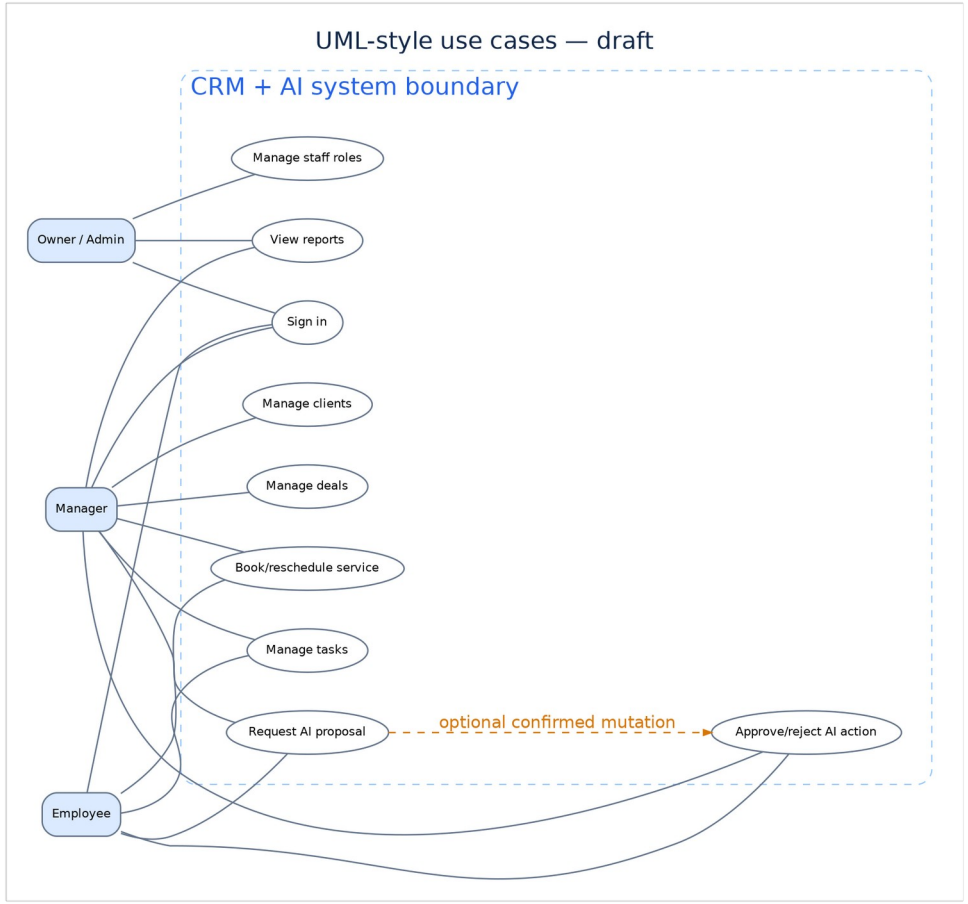
System context: users and external boundary

Human users and model provider do not bypass the application API



Use cases: human-owned workflows

Editable UML (.puml) and Graphviz sources are included



Owner / Admin

Access and governance

Manager / Employee

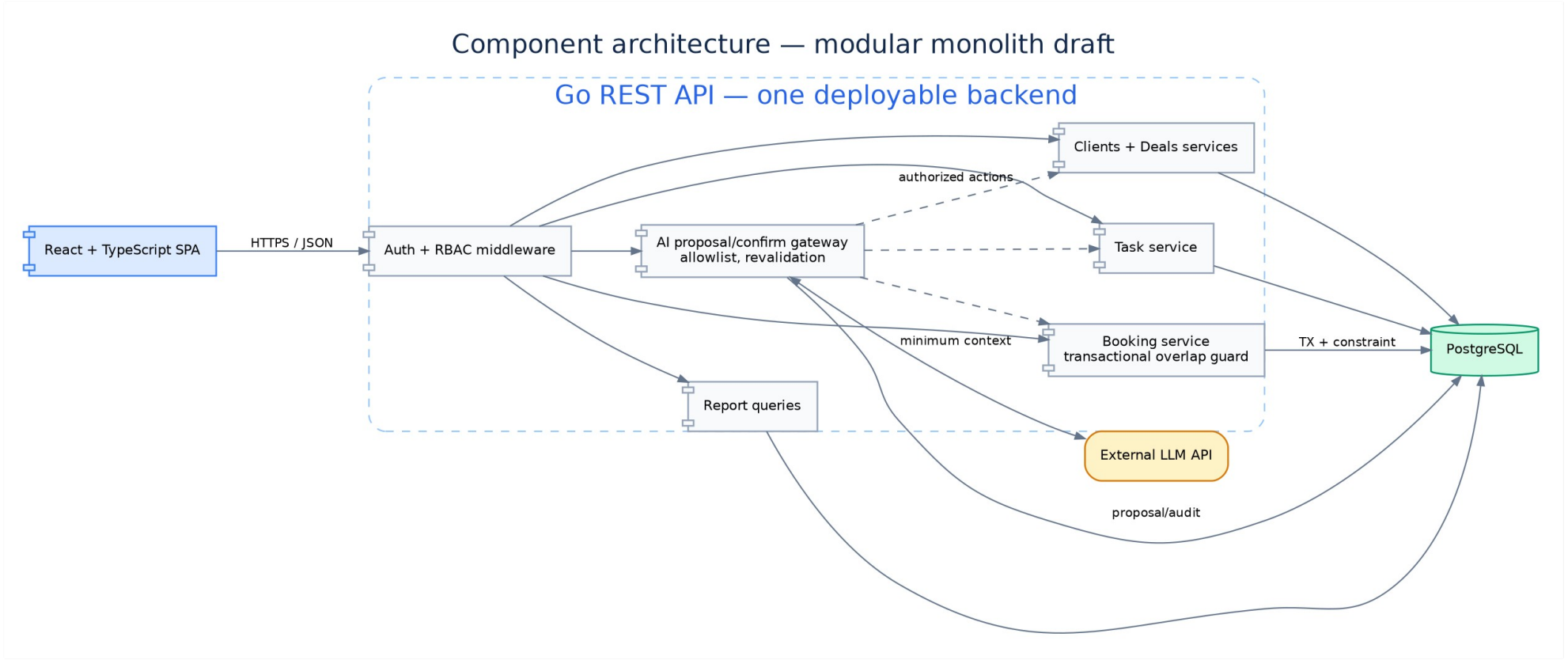
CRM, tasks and bookings

AI Assistant

Proposal only, not authorization

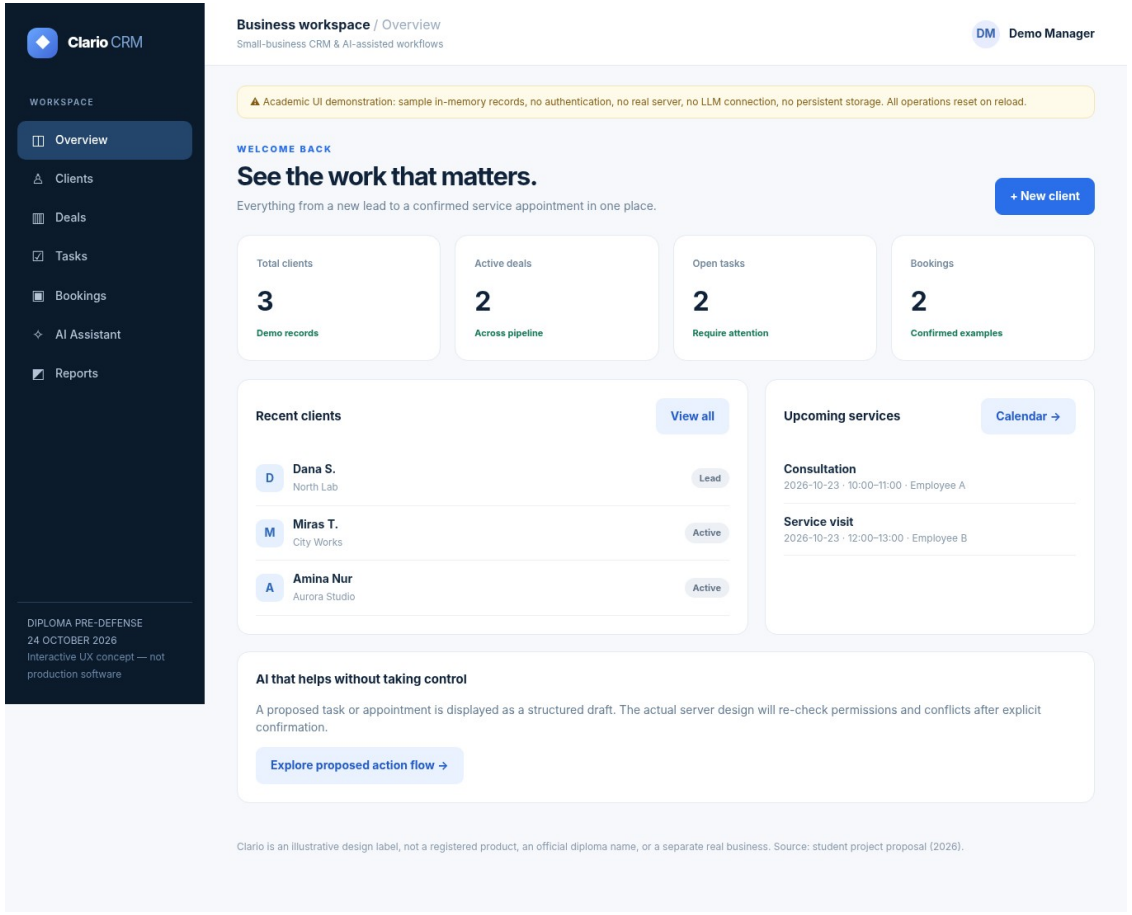
Modular monolith with guarded AI integration

One API, shared business rules and a transactional PostgreSQL persistence layer



A clickable prototype demonstrates key journeys

Standalone browser demo, using synthetic records and local in-memory state



Navigate

Overview, clients, deals, tasks, bookings

Create demo records

Inputs and tables update locally

AI draft

Visible proposal requires user approval

AI is an assistant, not an administrator

The language model never receives database credentials or general-purpose execution rights

01

Intent

User asks for action

02

Draft

Structured typed proposal

03

Review

User inspects changes

04

Gate

Auth + schema + invariants

05

Commit

Approved state mutation

Any failed permission, invalid data or collision → REJECT; no write

Evaluation compares outcomes—not promises

All thresholds are proposed targets; there are no real test results yet

0	0	≥85%	≥70	≥20%
overlapping confirmed bookings	**unauthorized state mutations**	**correct AI proposals**	**SUS score target**	**median time improvement target**
Concurrency test	Negative-role & AI corpus	100 annotated requests	Guided usability sessions	Against recorded manual baseline

Baseline: actual customer workflow or documented separate spreadsheet + calendar process; record setup and timing.

Team accountability and project workflow

Every meaningful task belongs to an owner, has acceptance criteria and links to Git evidence

A Aitzhan Iliyas

Backend + AI

API, schema, data invariants, guarded actions

A Aibek Galmbekov

Frontend

Views, UX flows, validation and integration

D Daniil Khvan

QA

Test strategy, cases, defect and safety checks

YouTrack task → feature branch → commit / PR → review → acceptance → tag

This process is prepared as a plan; real URLs, commit history, reviews and approvals still need to be supplied.

SUBMISSION STATUS

Pre-defense readiness: prepared vs outstanding

Transparent evidence tracking is required for an academically defensible submission

PREPARED IN THE PACKAGE

- ✓ Research review: 13 academic sources
- ✓ 4 competitors × 9 criteria
- ✓ 5 technology decisions × 3 options
- ✓ Requirements, 3 diagrams and prototype
- ✓ Test plan, risks and backlog templates

NEEDS REAL PARTICIPATION

- ! 3–5 real user interviews or client signoff
- ! Supervisor review and change records
- ! Repository and actual YouTrack project links
- ! Independent team commits and code reviews
- ! Reviewed first-stage Git tag

Thank you

Questions & feedback

Development of a Business Management and CRM Platform
with an Embedded AI Assistant